

Dheeraj Karanam

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SUMMARY

Data Engineer with production experience building and operating large-scale data pipelines and platforms in enterprise banking environments at IBM and TCS. Track record of delivering reliable, well-tested data systems from pipeline engineering and data modeling to quality frameworks and zero-downtime migrations. Currently completing an MS in Data Science at Indiana University.

WORK EXPERIENCE

IBM – Barclays Retail Banking

Hyderabad, India

Data Engineer

Mar 2023 – Aug 2025

- Designed and deployed **scalable data quality frameworks** (schema drift detection, null validation, outlier checks, row-count reconciliation) for large-scale distributed pipelines using Python and SQL; reduced debugging effort by **80%** and established monitoring standards adopted across **5** teams.
- Engineered and optimized **batch data pipelines** on **Hadoop/Spark** with Ab Initio, processing multi-million record retail banking datasets; built scalable data models and Data Lake architectures with end-to-end data lineage enabling downstream analytics and KPI forecasting for **5+** analytics teams.
- Led zero-downtime **platform migration** of **108 workflows** and **448 ETL jobs** (Cloudera CDH → BDH, RHEL6 → RHEL8) using CI/CD pipelines; improved system performance by **25%** and laid the foundation for cloud-native AWS migration.
- Established data contracts and schema evolution strategies across **3** business units; automated data movement and Hadoop file operations using Python and Unix scripts, reducing manual effort by **75%** and eliminating human errors in data movement operations.
- Drove performance optimization through systematic root-cause analysis, Spark job tuning, and code refactoring; collaborated with cross-functional teams in Agile environments (Jira) to deliver production-grade data systems on time.
- Maintained technical documentation, participated in code reviews, and presented pipeline architecture and data quality findings to both technical and non-technical stakeholders across business units.

TCS – Virgin Media

Chennai, India

Associate Software Engineer

Jul 2019 – Feb 2023

- Architected SCD Type 1 and 2 temporal data models for Netezza-to-GCP BigQuery Lakehouse migration across **645 tables**; applied schema evolution principles ensuring backward compatibility and zero data loss during production cutover using Ab Initio ETL.
- Optimized Apache Spark SQL transformations (partition pruning, predicate pushdown, partitioning and clustering rewrites) on GCP BigQuery, compressing a 14-hour batch pipeline to **12 hours**; documented execution plan analysis and bottleneck findings for engineering team.
- Maintained **99% SLA** and **99.5% uptime** across batch and near-real-time pipelines; recovered **50+ failed critical jobs** in Control-M by implementing error-handling mechanisms and monitoring protocols to ensure continuity.
- Delivered real-time data ETL applications using Informatica PowerCenter; resolved BI reporting issues across batch and real-time Ab Initio environments, ensuring **99% report delivery SLA** for critical business operations.
- Built Tableau and Power BI dashboards with interactive visualizations translating complex pipeline metrics into actionable insights for non-technical business leadership.
- Achieved **90% F1 accuracy** on NLP chatbot (Rasa NLU/Core) with REST API integration on AWS; deployed on production and presented at client town hall, driving adoption across 2 additional teams.
- Trained and guided **6 junior associates** in data warehouse concepts, ETL workflow design, and production support procedures, establishing knowledge transfer protocols for consistent project execution.

EDUCATION

Indiana University Bloomington

Aug 2025 – May 2027

M.S. in Data Science, Luddy School of Informatics, Computing, and Engineering

Andhra University, Visakhapatnam

Aug 2015 – Apr 2019

B.Tech, Electronics and Communication Engineering

CERTIFICATIONS

MITx ML with Python (2024) | MITx Statistics (2024) | MITx Probability (2024) | IBM Python for DS (2023)

TECHNICAL SKILLS

Data Engineering: Apache Spark/PySpark, Temporal Data Modeling (SCD Type 1/2), Schema Design and Evolution, Data Quality and Reconciliation, Batch and Streaming Pipelines, ETL/ELT, Data Lineage, Airflow

ETL Tools and DBs: Ab Initio, Informatica, Oracle, Teradata, Netezza, Hive, Impala, HDFS

Platforms and Cloud: Databricks (Delta Lake, Lakehouse), Apache Kafka, Apache Iceberg, Snowflake, GCP BigQuery, AWS

Data Formats: Parquet, Avro, JSON

Engineering: CI/CD, Git, Docker, Linux/Unix, Control-M, Jira, Agile, Confluence

Programming: Python (Pandas, NumPy, SciPy, FastAPI), SQL, R

ML and AI: Scikit-learn, XGBoost, PyTorch, RAG, LoRA, LLM APIs, HuggingFace

PROJECTS

Risk Modeling and Portfolio Analysis

[Git](#) | [Code](#)

- Benchmarked **6** VaR methods (regression, Monte Carlo, GARCH time series) on a 10-asset equity portfolio over **1,255** trading days; implemented data quality validation and reconciliation checks across all model outputs; validated with Kupiec hypothesis tests.
- Exposed a **\$3,300 per-\$100k risk gap** and **5.2x** volatility range between static and dynamic models; delivered structured findings with actionable insights for risk decision-making.

Financial Insight System – RAG, LoRA, MCP

[Git](#) | [Code](#)

- Built end-to-end data pipeline ingesting and parsing financial documents (JSON, PDF); implemented schema-aligned extraction, confidence scoring, and data quality validation across SEC filings via RAG retrieval and LoRA fine-tuned LLM.
- Developed FastAPI backend with evaluation dashboard tracking retrieval accuracy and output quality metrics (BERTScore, F1-score); delivered with runbook-style documentation.

CAG vs RAG Evaluation Framework – Medical Genomics

Prof. Madurima, IU

- Designed and built an end-to-end evaluation pipeline comparing three retrieval architectures (RAG, CAG, CRAG) for NLP-based information extraction from medical genomics documents; implemented confidence scoring, hallucination rate measurement, and automated validation checks across transformer-based models.
- Executed pipeline on BigRed200 HPC (SLURM, A100 GPUs) using LangChain, FAISS, and HuggingFace Transformers; structured as a reproducible benchmarking framework with consistent data quality controls across all model outputs.

Self-Supervised Representation Learning – ResNet, CLTT

[Git](#) | [Code](#)

- Built and operated a large-scale deep learning training pipeline (ResNet-18, contrastive loss) processing **50K+ video frames** on BigRed200 HPC (A100 GPUs, SLURM); designed systematic ablation framework varying batch size, learning rate, and augmentation to evaluate representation quality.
- Implemented kNN retrieval and linear probe evaluation harness to measure downstream pipeline output quality across all experimental configurations.